

TEEJARTI Platform - Comprehensive Go Live and Testing Plan

System Under Test (SUT) Analysis

TEEJARTI is a full-stack business matchmaking platform for Oman and GCC region with:

- **Frontend:** React 18 + TypeScript, i18next (English/Arabic), Tailwind CSS
- **Backend:** Node.js + Express, PostgreSQL + Drizzle ORM, JWT authentication
- **Key Features:** Business listings, investor matching, KYC verification, admin management, bilingual support

Risk Assessment Matrix

Risk Level	Component	Impact	Likelihood	Mitigation Priority
HIGH	Authentication System	High	Medium	1
HIGH	Payment Processing	High	Low	2
HIGH	Data Security/Privacy	High	Medium	3
MEDIUM	KYC Verification	Medium	Medium	4
MEDIUM	Translation Services	Medium	High	5
LOW	UI/UX Responsiveness	Low	Low	6

Comprehensive Test Plan

1. FUNCTIONAL TESTING

1.1 Authentication & Authorization Module

Test Case ID	Test Case	Type	Priority	Expected Result
AUTH_001	User Registration with Valid Data	Positive	High	User created, verification email sent
AUTH_002	Registration with Duplicate Email	Negative	High	Error: "Email already registered"
AUTH_003	Registration with Invalid Email Format	Negative	High	Validation error returned
AUTH_004	Login with Valid Credentials	Positive	High	JWT token generated, user authenticated

Test Case ID	Test Case	Type	Priority	Expected Result
AUTH_005	Login with Invalid Credentials	Negative	High	Error: "Invalid credentials"
AUTH_006	JWT Token Expiry (30 days)	Edge	Medium	Token expires, re-authentication required
AUTH_007	Password Reset Flow	Positive	Medium	Reset email sent, password updated
AUTH_008	Role-based Access (Admin vs Entrepreneur)	Authorization	High	Admin-only routes protected
AUTH_009	Rate Limiting on Login (5 attempts/min)	Security	High	Requests blocked after limit
AUTH_010	Email Verification OTP	Positive	Medium	Account verified with valid OTP

Test Data Requirements:

```
{
  "validUser": {
    "username": "testuser001",
    "email": "test@teejarti-qa.com",
    "password": "TestPass123!",
    "fullName": "Test User",
    "role": "entrepreneur"
  },
  "invalidEmails": [
    "invalid-email", "@domain.com", "test@", ""
  ]
}
```

1.2 Business Listings Module

Test Case ID	Test Case	Type	Priority	Expected Result
LIST_001	Create Business Listing (Complete Data)	Positive	High	Listing created with "pending" status

Test Case ID	Test Case	Type	Priority	Expected Result
LIST_002	Create Listing with Missing Required Fields	Negative	High	Validation errors for required fields
LIST_003	Upload Business Images (Valid Formats)	Positive	Medium	Images uploaded, paths stored
LIST_004	Upload Invalid File Format	Negative	Medium	Error: "Only image files allowed"
LIST_005	Search Listings by Location	Functional	Medium	Filtered results by location
LIST_006	Search Listings by Industry	Functional	Medium	Filtered results by industry
LIST_007	Listing Approval Workflow (Admin)	Workflow	High	Status changes from "pending" to "approved"
LIST_008	Listing Rejection Workflow (Admin)	Workflow	High	Status changes to "rejected" with reason
LIST_009	Edit Own Listing (Owner)	Positive	Medium	Listing updated successfully
LIST_010	Edit Other User's Listing (Unauthorized)	Security	High	Access denied
LIST_011	Bilingual Content (English/Arabic)	I18n	Medium	Content displayed in selected language
LIST_012	View Listing Details (Public)	Positive	Medium	Complete listing information displayed

1.3 KYC Verification Module

Test Case ID	Test Case	Type	Priority	Expected Result
KYC_001	Submit KYC Documents (All Required)	Positive	High	KYC application created, status "pending"
KYC_002	Submit Partial KYC Documents	Negative	High	Error: Missing required documents

Test Case ID	Test Case	Type	Priority	Expected Result
KYC_003	Upload Invalid Document Format	Negative	Medium	Error: "Invalid document format"
KYC_004	Admin Approve KYC Application	Workflow	High	Status "approved", user verified
KYC_005	Admin Reject KYC Application	Workflow	High	Status "rejected" with reason
KYC_006	Resubmit Rejected KYC	Edge	Medium	New application created
KYC_007	Document File Size Limits (10MB)	Edge	Medium	Large files rejected

2. API TESTING

2.1 REST API Endpoints Testing

Endpoint	Method	Test Cases	Auth Required	Expected Status
/api/auth/register	POST	Valid/Invalid data, duplicates	No	201/400
/api/auth/login	POST	Valid/Invalid credentials	No	200/401
/api/listings	GET	Filtering, pagination	No	200
/api/listings	POST	Create listing	Yes	201/401
/api/listings/:id	GET	Valid/Invalid ID	No	200/404
/api/admin/listings	GET	Admin access only	Yes (Admin)	200/403
/api/kyc	POST	Submit KYC docs	Yes	201/401
/api/contacts	POST	Contact seller	Yes	201/401
/api/settings	GET	Public settings	No	200
/api/admin/settings	PATCH	Update settings	Yes (Admin)	200/403

2.2 API Security Testing

Test Case	Description	Expected Result
SQL Injection	Inject SQL in request parameters	Sanitized by Drizzle ORM
XSS Attack	Script injection in form fields	Sanitized by validation
CSRF Protection	Requests without proper headers	Blocked by middleware
Rate Limiting	Exceed request limits	429 Too Many Requests
JWT Manipulation	Modified/expired tokens	401 Unauthorized

3. DATABASE TESTING

3.1 Data Integrity Tests

Test Area	Test Cases
Foreign Key Constraints	Verify referential integrity between users-listings-kyc
Data Validation	Database-level constraints enforce data rules
Transaction Rollback	Failed operations don't leave partial data
Concurrent Access	Multiple users creating listings simultaneously
Data Backup/Recovery	Database backup and restore procedures

3.2 Database Schema Validation

-- Test critical foreign key relationships

```
SELECT COUNT(*) FROM listings l
```

```
LEFT JOIN users u ON l.userId = u.id
```

```
WHERE u.id IS NULL;
```

-- Expected: 0 (no orphaned listings)

-- Test data consistency

```
SELECT role, COUNT(*) FROM users
```

```
WHERE role IN ('entrepreneur', 'investor', 'broker', 'admin')
```

```
GROUP BY role;
```

-- Expected: All users have valid roles

4. INTERNATIONALIZATION (I18N) TESTING

Test Case	Description	Languages	Expected Result
UI Language Switch	Toggle between English/Arabic	EN/AR	Interface updates completely
RTL Layout	Arabic text direction	AR	Right-to-left layout applied
Translation Completeness	All UI elements translated	AR	No English text in Arabic mode
Date/Time Formatting	Locale-specific formats	EN/AR	Correct formatting per locale
Currency Display	OMR currency formatting	EN/AR	Proper currency symbols
Translation Keys	Missing translation keys	EN/AR	No raw keys displayed

5. SECURITY TESTING

5.1 Authentication Security

Test Case	Description	Expected Result
Password Hashing	Passwords stored using bcrypt	Hashed passwords in database
JWT Security	Token signing and verification	Secure token generation
Session Management	User session handling	Proper session lifecycle
Two-Factor Authentication	OTP verification flow	Additional security layer

5.2 File Upload Security

Test Case	Description	Expected Result
File Type Validation	Upload non-image files	Rejected
File Size Limits	Upload oversized files	Rejected
File Name Sanitization	Upload files with dangerous names	Names sanitized
Directory Traversal	Upload attempts with ../ paths	Blocked

6. PERFORMANCE TESTING

6.1 Load Testing Scenarios

Scenario	Users	Duration	Expected Response
Normal Load	10 concurrent	5 minutes	< 2s response time
Peak Load	50 concurrent	10 minutes	< 5s response time
Stress Test	100 concurrent	15 minutes	Graceful degradation
Database Load	1000 listings query	-	< 3s query time

6.2 Performance Benchmarks

API Response Time Testing

```
curl -w "@curl-format.txt" -s -o /dev/null https://api.teejarti.com/api/listings
```

Target: < 1000ms

Database Query Performance

```
EXPLAIN ANALYZE SELECT * FROM listings
```

```
WHERE status = 'approved'
```

```
ORDER BY createdAt DESC LIMIT 20;
```

Target: < 100ms execution time

7. USABILITY TESTING

7.1 User Experience Flows

User Journey	Steps	Success Criteria
New User Registration	Register → Verify Email → Complete Profile → Submit KYC	< 10 minutes completion
Create Business Listing	Login → Create Listing → Upload Images → Submit	< 5 minutes completion
Browse Listings	Search → Filter → View Details → Contact Seller	< 3 minutes to contact
Admin Moderation	Login → Review Listings → Approve/Reject	< 2 minutes per listing

7.2 Accessibility Testing

Test Area	Standard	Tools
WCAG Compliance	WCAG 2.1 AA	axe-core, WAVE
Screen Reader	NVDA/JAWS compatibility	Manual testing
Keyboard Navigation	Tab order, focus management	Manual testing
Color Contrast	4.5:1 ratio minimum	Color Contrast Analyzer

8. MOBILE RESPONSIVENESS TESTING

Device Category	Test Devices	Screen Sizes
Mobile	iPhone 12, Galaxy S21	375px, 414px
Tablet	iPad, Galaxy Tab	768px, 1024px
Desktop	Chrome, Safari, Firefox	1200px, 1920px

9. CROSS-BROWSER TESTING

Browser	Version	OS	Priority
Chrome	Latest	Windows/Mac	High
Firefox	Latest	Windows/Mac	High
Safari	Latest	Mac/iOS	Medium
Edge	Latest	Windows	Medium

Traceability Matrix

Requirement ID	Requirement Description	Test Case IDs	Coverage Status
REQ_001	User Registration & Authentication	AUTH_001-010	✓ Complete
REQ_002	Business Listing Management	LIST_001-012	✓ Complete
REQ_003	KYC Verification Process	KYC_001-007	✓ Complete
REQ_004	Admin Management Panel	ADM_001-008	⚠ Partial
REQ_005	Bilingual Support (EN/AR)	I18N_001-006	✓ Complete
REQ_006	Email Notifications	EMAIL_001-005	⚠ Partial
REQ_007	File Upload & Management	FILE_001-006	✓ Complete
REQ_008	Contact Seller Feature	CONTACT_001-004	✗ Missing
REQ_009	Payment Integration	PAY_001-005	✗ Missing
REQ_010	Analytics & Reporting	ANALYTICS_001-003	✗ Missing

Coverage Summary:

- **Complete Coverage:** 60% (6/10)
- **Partial Coverage:** 20% (2/10)
- **Missing Coverage:** 20% (2/10)

Test Environment Setup

Development Environment

Database setup

npm run db:push

Environment variables

DATABASE_URL="postgresql://..."

JWT_SECRET="your-secret-key"

SENDGRID_API_KEY="your-sendgrid-key"

Start application

npm run dev

Test Data Generation

-- Create test users

```
INSERT INTO users (username, email, password, fullName, role) VALUES
('testadmin', 'admin@teejarti-qa.com', '$hashed_password', 'Test Admin', 'admin'),
('testuser1', 'user1@teejarti-qa.com', '$hashed_password', 'Test User 1', 'entrepreneur'),
('testinvestor', 'investor@teejarti-qa.com', '$hashed_password', 'Test Investor', 'investor');
```

-- Create test listings

```
INSERT INTO listings (userId, title, description, industry, location, askingPrice, status) VALUES
(2, 'Test Business 1', 'Description', 'Technology', 'Muscat', 50000, 'approved'),
(2, 'Test Business 2', 'Description', 'Manufacturing', 'Dubai', 100000, 'pending');
```

Automation Recommendations

1. Unit Testing Framework

```
npm install --save-dev jest @types/jest supertest
```

```
npm install --save-dev @testing-library/react @testing-library/jest-dom
```

2. API Testing Automation

// Example API test with Jest/Supertest

```
describe('Authentication API', () => {
  test('POST /api/auth/register should create user', async () => {
    const response = await request(app)
      .post('/api/auth/register')
      .send(testUser)
      .expect(201);

    expect(response.body.user.email).toBe(testUser.email);
  });
});
```

3. E2E Testing with Playwright

// Example E2E test

```
test('Complete user registration flow', async ({ page }) => {
  await page.goto('/auth');
  await page.fill('[name="email"]', 'test@example.com');
  await page.fill('[name="password"]', 'password123');
  await page.click('button[type="submit"]');
  await expect(page.locator('.success-message')).toBeVisible();
});
```

```
4. CI/CD Integration

# GitHub Actions workflow

name: Test Suite

on: [push, pull_request]

jobs:

  test:
```

```
    runs-on: ubuntu-latest

    steps:
      - uses: actions/checkout@v3
      - uses: actions/setup-node@v3
      - run: npm ci
      - run: npm run test
      - run: npm run e2e
```

Monitoring & Reporting

Test Metrics to Track

- **Test Coverage:** Aim for >80% code coverage
- **API Response Times:** <2s for critical endpoints
- **Error Rate:** <1% for production
- **User Flow Completion:** >95% success rate

Reporting Tools

- **Jest Coverage Reports:** HTML/JSON output
- **Playwright Reports:** Test results with screenshots
- **Performance Monitoring:** Response time tracking
- **Error Tracking:** Sentry/LogRocket integration

Risk Mitigation Strategies

Risk	Impact	Mitigation Strategy
Authentication Bypass	High	Implement comprehensive auth testing, security code reviews
Data Loss	High	Regular backups, transaction testing, rollback procedures
Performance Degradation	Medium	Load testing, monitoring, caching strategies
Translation Errors	Medium	Native speaker review, automated translation validation

Risk	Impact	Mitigation Strategy
Mobile Compatibility	Low	Responsive design testing, device-specific testing

Recommendations for Implementation

Phase 1 (Immediate - Week 1)

- 1. Set up basic unit testing framework
- 2. Create API endpoint tests for authentication
- 3. Implement database integrity tests

Phase 2 (Short-term - Week 2)

- 1. Expand API test coverage to all endpoints
- 2. Implement E2E tests for critical user journeys
- 3. Add performance benchmarking
- 4. Create test data management system

Phase 3 (Week 3-4)

- 1. Implement security testing automation
- 2. Add comprehensive mobile testing
- 3. Set up monitoring and alerting
- 4. Create test reporting dashboards

This comprehensive test plan provides structured coverage of all critical system components with emphasis on risk-based testing prioritization. The traceability matrix ensures no requirements are missed, while the automation recommendations provide a clear path for implementation.